

Muhammad Mahwiz Khalil

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SUMMARY

Python engineer and NLP/ML researcher specializing in deep learning systems and AI infrastructure for low-resource languages, with a focus on Urdu speech and text technologies. Experienced in model training, fine-tuning, and distillation, GPU inference serving, and cloud deployment (AWS EC2, Docker, FastAPI, nginx). Maintains an active open-source research portfolio of 40+ HuggingFace models and 20+ datasets.

CORE TECHNICAL COMPETENCIES

- **Programming & Architecture:** Python, C, SQL, System Design, Distributed Computing
- **Backend Engineering:** Flask, Django, FastAPI, Microservices Architecture, RESTful APIs
- **Machine Learning Platform:** PyTorch, TensorFlow, HuggingFace Ecosystem, MLOps, Model Deployment
- **NLP & Speech Systems:** Transformers, Custom Model Development, TTS/ASR, Grapheme-to-Phoneme, Model Distillation, ONNX
- **Data Engineering:** PostgreSQL, MySQL, ChromaDB, FAISS, Vector Databases, ETL Pipelines
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, nginx, Kubernetes, CI/CD, GPU Inference Serving, Production Monitoring
- **Development Ecosystem:** Git, Linux, Agile Methodologies, Code Review, Technical Leadership

PROFESSIONAL EXPERIENCE

NLP Engineer *Proxima AI — Pakistan* **August 2023 – Present**

- Engineered a chatbot platform, achieving a 50% reduction in response latency
- Built production-ready Speech-to-Text and Text-to-Speech systems for Urdu language processing, including the Aegis, Cadenza, DeepSpeak-v1, and Kani-TTS Urdu TTS models
- Led development of ATUP, an asynchronous Urdu OCR/TTS pipeline; improved Nastaliq glyph recognition accuracy from 34% to 93% and reduced first-response latency to approximately 0.04 seconds
- Developed a scalable image generation pipeline using Stable Diffusion, processing over 10,000 requests daily
- Established a technical mentorship program, resulting in a 30% improvement in team delivery metrics
- Optimized deployment workflows on AWS EC2 and Docker, reducing infrastructure costs by 25% while improving system reliability

Visiting Faculty *Educad Academy — Pakistan* **2025 – Present**

- Designed and delivered industry facing courses across Agentic AI, Machine Learning, Deep Learning, RFID Systems, AI for Finance and Statisticals.
- Authored professional course brochures and curriculum materials for multiple subject areas
- Mentored working professionals and students transitioning into applied AI roles

AI Engineer *Nova Tech — Dubai, UAE (Remote)* **June 2025 – June 2026**

- Developed an AI agent-based cashflow management copilot system
- Architected an intelligent trading bot using real-time market data analysis, achieving 85% accuracy with an AI agent
- Established a company-wide AI governance policy framework ensuring compliance with UAE financial regulations
- Led implementation of multi-agent systems for portfolio management, risk assessment, and automated decision-making

Data Research Analyst *Technexia — Pakistan* **January 2023 – June 2023**

- Developed automated data intelligence systems using Python, processing over 1 million data points monthly
- Built scalable web scraping infrastructure with rate limiting and error handling mechanisms
- Created executive dashboards and analytical reports to support strategic business decisions

RESEARCH & PUBLICATIONS

- “Direct Preference Optimization for Urdu: Translating HH-RLHF and Fine-Tuning Qalb-1.0-8B” — submitted to MRL 2026 (EMNLP Workshop)
- Contributor, multiple peer-reviewed and community publications in multilingual and low-resource NLP

KEY ENGINEERING PROJECTS

URNOVA-95M *ModernBERT, Apache-2.0, 8,192-token context* **2026**

- Built an Urdu-first ModernBERT encoder and benchmarked it on the POLAR binary polarization task

DeepSpeak-v1 *Rectified Flow DiT, Qwen3.5-0.8B, Semantic DACVAE* **2026**

- Trained an Urdu text-to-speech model from scratch using rectified flow matching with a Qwen3.5-0.8B text backbone and reference-audio speaker conditioning, supporting voice cloning from 3–15 second clips

Aegis TTS *OmniVoice Distillation, Piper, ONNX* **2025**

- Distilled a 61MB CPU-inference female Urdu TTS model from OmniVoice, trained on over 32,000 filtered synthetic speech samples; contributed the voice to the Rhasspy piper-voices repository

Urdu Legal Privacy Filter *Bidirectional Transformer, BIOES / Viterbi Decoding* **2025**

- Fine-tuned a PII detection and redaction model for Pakistani legal Urdu documents (court orders, FIRs, judgments) across six entity types, trained on 17,360 annotated windows

Lafzyn *Qwen3.5-0.8B* **2025**

- Built an Urdu grapheme-to-phoneme model converting Urdu text to IPA, released with a live inference demo

LittleMarkhor *DeepSeek-V4 MoE Architecture, FineWeb-2, Aya Collection* **2025**

- Pretrained a 110-million-parameter Urdu language model from scratch and applied supervised fine-tuning for instruction following

Kani-TTS-400M-UR *Kani-TTS 400M* **2025**

- Released the smallest known Urdu TTS model, optimized for low-resource language deployment conditions

TinyGemma-Urdu / Whisper-Urdu-PyTorch / Urdu-BERT *PyTorch, Custom Tokenization* **2024 – 2025**

- Implemented Gemma (0.96M parameters), Whisper (135M parameters), and BERT architectures from scratch, and pretrained each on Urdu text and speech corpora, including a custom WordPiece tokenizer

EDUCATION

Bachelor of Science in Computer Science **September 2024 – August 2027**
University of Karachi, Karachi, Pakistan

Diploma in Computer Information Technology **October 2020 – October 2023**
Aligarh Institute of Technology, Pakistan

PROFESSIONAL CERTIFICATIONS & INDUSTRY RECOGNITION

- Google Cloud — Advanced Encoder-Decoder Architecture & Transformer Models Specialization
- AWS — Practical Data Science on AWS Cloud Specialization with Production ML Focus
- IBM — Comprehensive Data Science Professional Certificate with Deep Learning Concentration
- Kaggle Notebook Expert — Top 1% contributor with 50,000+ views on machine learning tutorials
- Distinguished Technical Speaker: “Advanced HuggingFace Transformers and LLMs” at SMIU Conference
- Open Source Maintainer: HuggingFace handle mahwizzzz / GitHub mwzkhalil — 40+ production-ready models, 20+ datasets